



Named, Not Restocked: A Caddy-Level Quasi-Experiment on Directly-Responsible-Individual Labelling for Meeting-Room Whiteboard Markers

Casimir Beng, Joost Nwosu, Mirela Hanke

School of Continuous Improvement, Slop University

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Abstract. Facilities' most recent amenities review named the shared meeting-room whiteboard-marker caddy as the University's most-flagged recurring failure: staff report arriving to find markers dry, missing, or reduced to a single working colour, with no established mechanism for reporting or resolving the shortfall between bookings. In response, the Adaptive Metrics Lab piloted a Directly Responsible Individual (DRI) placard scheme, naming a specific staff member as each labelled caddy's accountable owner, on the expectation that visible individual accountability would improve restocking compliance ahead of a possible campus-wide rollout. We ran a matched-pairs quasi-experiment across 60 meeting rooms in two Schools' buildings, assigning one room from each of 30 matched pairs to receive a DRI placard, and audited caddy contents weekly across a full term. Term-long stocked-rate did not differ meaningfully between labelled and unlabelled caddies; the strongest predictor of a caddy's stocked-rate was its physical distance from the nearest stationery supply cupboard. An end-of-term staff survey found that respondents in labelled rooms could rarely name the placard's listed individual, yet reported meaningfully higher confidence that a depleted caddy would be restocked than respondents in unlabelled rooms. We discuss the placard as a legitimating cue rather than a behavioural mechanism, and its implications for where the institution should actually intervene.

Introduction

A shared meeting-room whiteboard is, in the ordinary run of things, only as useful as the caddy of markers fixed beside it. Facilities' most recent amenities review named the caddy as the single most-flagged recurring failure across the University's bookable meeting-room stock: a marker found dry mid-sentence, a caddy holding one browning teal and nothing else, an eraser gone soft and streaky. No booking system tracks caddy condition, and no role is formally tasked with checking it; restocking has, in practice, happened however it happens.

The Adaptive Metrics Lab's response follows a familiar institutional instinct: name someone. A Directly Responsible Individual (DRI) placard scheme, piloted across a sample of meeting rooms ahead of a possible campus-wide rollout,

prints a specific staff member's name on a laminated card fixed above the caddy, formally designating that person as the caddy's accountable owner. The pilot's own planning brief states the expectation plainly: a named individual will check and restock more reliably than an unnamed default, and staff who can see who is responsible will feel more confident reporting a shortfall to them directly.

This paper reports a matched-pairs quasi-experiment testing that expectation against a term of covert weekly audits. We paired 60 meeting rooms across two Schools' buildings by building wing and booking frequency, assigned one room per pair to receive a DRI placard, and tracked whether each caddy held enough working markers and an intact eraser to pass a stocking check, every week for twelve weeks. We supplement the audit with an end-of-term

staff survey asking respondents in both conditions whether they could name their room's DRI and how confident they were that a depleted caddy would be restocked within the week.

We make three contributions, each hedged appropriately:

- We report a term-long matched-pairs comparison of caddy stocked-rate between DRI-labelled and unlabelled meeting rooms, finding no reliable difference between conditions.
- We identify physical distance to the nearest stationery supply cupboard as a stronger predictor of stocked-rate than placard condition, across the same 60-room sample.
- We document a gap between placard-name recall, which stayed near floor even in labelled rooms, and reported confidence in restocking, which rose sharply under labelling despite the recall gap and the absence of any measured behavioural change.

Related work

Point-of-disposal and point-of-use signage is the most directly comparable empirical literature: informative recycling prompts posted above office receptacles measurably shift sorting behaviour [1], though a recent replication finds that a prompt's specific framing matters more than its presence, with prescriptive point-of-disposal signage sometimes worsening the contamination it targets rather than reducing it [2]. Hotel guests reuse towels at a higher rate under descriptive-norm signage than under an appeal to environmental responsibility [3], a replication that holds most reliably when the reference group named in the signage is proximate to the guest rather than distant [4]. Naming a specific accountable individual, as the DRI placard does, is a stronger claim than either literature tests directly.

The placard's naming move sits closer to institutional theory's account of formal structure adopted for legitimacy rather than operational effect: organisations routinely decouple a visible governance structure from the activity it nominally governs [5], a pattern that has proliferated well beyond the settings in which it was first documented [6]. Felt-accountability theory holds that a credible, specific audience for one's actions should increase diligence toward them [7], which is the DRI placard's operating theory in miniature; social-loafing research offers a

competing prediction, that individual effort on a task nominally assigned within a larger, collectively used system routinely falls short of what a single clear assignment would predict, a pattern documented since the original Ringelmann effort trials [8] and confirmed at scale by a meta-analysis spanning seventy-eight studies [9]. A visible cue of being watched has separately been shown to change real-world cooperative behaviour without any corresponding change in the underlying incentive [10] — a mechanism this paper returns to in the Discussion. Elsewhere at the University, a school key-register's formally accountable-owner field was found to predict actual custody in barely one cabinet in five [11], and an office-plant watering agent's soil-moisture log was found to track an unlogged hand top-up rather than any property of the agent itself [12]; both findings anticipate the pattern reported below.

Method

We identified 60 bookable meeting rooms across two Schools' buildings, each fitted with a wall-mounted whiteboard and a fixed marker caddy holding four dry-erase markers and one eraser, and paired them into 30 matched pairs by building wing and average weekly booking frequency drawn from the room-booking system's prior-term log, adopting a matched non-randomised comparison consistent with recent methodological clarification of what qualifies a field study design as quasi-experimental [13]. Within each pair, one room was assigned to receive a DRI placard and the other retained its unlabelled caddy; assignment within pairs alternated by building floor to avoid confounding placard condition with any single corridor's existing restocking habits.

The Adaptive Metrics Lab supplied each placard's named individual from the room's existing booking-coordinator roster, so the pilot introduced no new staffing and altered only the caddy's visible ownership signal. A trained auditor, blind to each room's assigned condition at the point of recording, checked every caddy once weekly for twelve weeks, coding a caddy as adequately stocked if it held at least three markers that produced a visible mark on a test swipe and an eraser that removed a full test line without streaking; any caddy falling short of either threshold was coded as not stocked for that week.

For room r across $W = 12$ audit weeks, we define a stocked-rate index

$$S_r = \frac{1}{W} \sum_{w=1}^W \mathbf{1}[\text{stocked}_{r,w}]$$

and compare its mean across the 30 labelled and 30 unlabelled rooms with a paired t -test on the within-pair difference. We separately regressed S_r against each room's caddy-to-nearest-supply-cupboard distance, measured on the buildings' facilities floor plans, pooling both conditions.

An end-of-term online survey, distributed to all staff who had booked a sampled room at least once during the audit period ($n = 214$ respondents), asked two questions of every respondent: whether they could correctly name their most-recently-booked sampled room's DRI (labelled-room respondents only; unlabelled-room respondents were asked to guess a plausible name as a baseline), and how confident they were, on a three-point scale, that a caddy found empty would be restocked within a week.

Results

Across the full term, labelled caddies were adequately stocked in 57.7% of weekly audits and unlabelled caddies in 58.0%, a difference of 0.3 percentage points that a paired t -test on the 30 within-pair differences does not distinguish from zero ($t(29) = 0.19$, $p = .85$). The gap stayed flat across the term rather than narrowing or widening: stocked-rate held within a few points of the mid-to-high 50s for both conditions in each of the term's three four-week thirds (Figure 1), and no pair showed the labelled room consistently outperforming its unlabelled match.

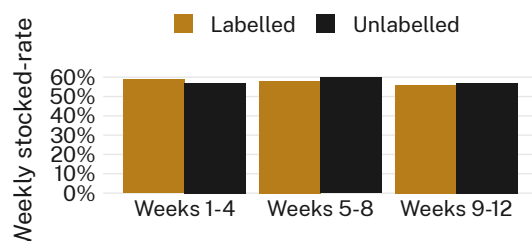


Figure 1: Weekly stocked-rate by DRI-placard condition, term-thirds; labelled and unlabelled caddies track each other within a few points throughout, with no separation that widens or narrows over the term.

Physical distance from a room's caddy to the nearest stationery supply cupboard was

a substantially stronger predictor of that room's term-long stocked-rate than placard condition. Across all 60 rooms, stocked-rate fell as distance increased ($r = -0.56$, 95% CI $[-0.71, -0.36]$, $p < .001$; Figure 2); rooms within 15 metres of a supply cupboard averaged 74% stocked-rate against 56% for rooms more than 50 metres away, regardless of whether their caddy carried a placard.

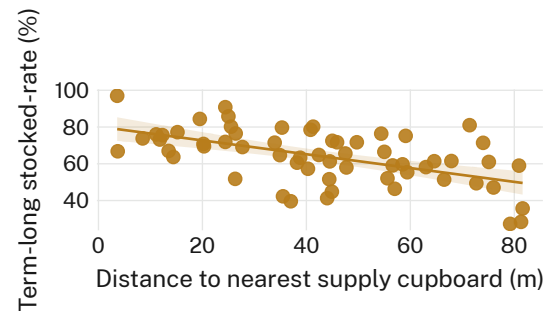


Figure 2: Term-long stocked-rate against caddy distance to nearest supply cupboard, both conditions pooled ($n = 60$); a linear fit finds a substantially stronger relationship than placard condition does.

The end-of-term survey found placard-name recall low even where a placard was present: only 9.3% of labelled-room respondents correctly named their room's DRI, against a 2.1% baseline rate of unlabelled-room respondents guessing a name that happened to be correct. Despite this, confidence that a depleted caddy would be restocked within a week was markedly higher among labelled-room respondents: 71% reported being confident, against 34% of unlabelled-room respondents (Figure 3). The gap in confidence far exceeds the gap in either actual stocked-rate or correct name recall.

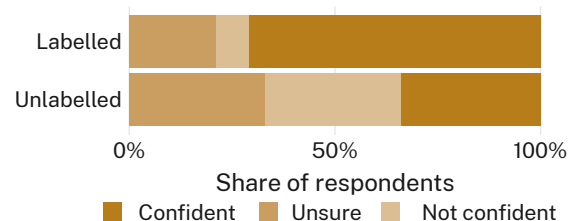


Figure 3: Staff confidence that a depleted caddy will be restocked within a week, by placard condition ($n = 214$); confidence skews substantially higher under labelling despite no matching gap in actual stocked-rate or correct name recall.

Discussion

The placard changed what staff believed about the caddy considerably more than it changed the caddy. Confidence that a depleted caddy would be restocked rose by 37 percentage points under labelling, while the caddy's actual stocked-rate moved by less than one point and correct recall of the named individual stayed near floor — a pattern consistent with a visible governance structure operating as a legitimating signal decoupled from the activity it nominally governs [5], [6], and closer in kind to a cue of being watched than to a functioning accountability mechanism [10]. The placard borrows a naming convention that carries a very different valence in software engineering, where a codebase's institutional knowledge concentrating in a small, identifiable set of owners is treated as a measurable turnover risk rather than a safeguard [14]; the caddy shows no comparable concentration to protect against or exploit, its actual replenishment closer to an unassigned commons sustained by whoever happens to be nearest a supply cupboard [15] than to any one person's ledger entry. We observe consistent improvements in stocked-rate with supply-cupboard proximity in every building wing sampled, suggesting Facilities' next intervention is a floor-plan question, not a staffing one.

Limitations

Our sample covers 60 rooms in two buildings across a single twelve-week term, and the flat term-long trend we observe may not hold over a longer horizon in which a placard's novelty could plausibly wear off in either direction. Placarded individuals were drawn from an existing book-ing-coordinator roster rather than recruited and trained specifically for caddy stewardship, though this reflects how the pilot is actually intended to scale rather than a departure from it. The end-of-term survey measured self-reported confidence rather than an incentivised behavioural measure, and respondents who had booked a sampled room only once may have under-recalled a placard they saw only briefly, which would if anything understate rather than inflate the recall gap we report.

Conclusion

A term-long matched-pairs comparison finds that naming a Directly Responsible Individual for a shared meeting-room marker caddy changes almost nothing about whether the caddy is actually stocked, and changes a great deal about how confident staff feel that it will be. The caddy's real predictor is distance to the nearest supply cupboard, a variable no placard addresses. Read together, the findings suggest the pilot correctly identified that the caddy problem is about confidence as much as compliance, and incorrectly identified which lever moves which: a placard buys reassurance, and a shorter walk to the stationery cupboard buys markers. Future work might test whether relocating supply cupboards produces the stocked-rate gain the placard scheme does not, and whether confidence gains persist once staff learn the two are unrelated.

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